Ubuntu va OS asoslari: to‘liq mashqlar

P0–P6: topshiriqlar va qadamma-qadam yechim

Ubuntu Server 24.04 LTS
JDU Cyber Security / 2026 / ChatGPT tarjima qoralamasi

Ubuntu and OS Foundations — to‘liq ko‘rsatmalar bilan mashqlar

> ChatGPT tarjimasi joriy yaponcha manbaga moslashtirilgan. Ona tilida gapiruvchi mutaxassis hali tekshirmagan; baholash shartlari farq qilsa, yaponcha asl matnga tayaning.

Qamrov: Guided Practice P0–P6. Muhit: Ubuntu Server 24.04 LTS.

Tartib: avval har bir P topshirig‘ini o‘qing. O‘zingiz yechib ko‘ring, so‘ng qadamma-qadam yechim bilan amallar va sabablarini solishtiring. Shu ko‘nikmaga tegishli Mission Mni mustaqil bajaring.

Avval Ubuntuga ulaning

AWS Academy Learner Lab dastlabki sozlamalari tugagach, AWS konsolida CloudShellni oching. CloudShellda o‘z Ubuntu mashq serveringizga ulaning:

```
ssh jdu-ubuntu
```

P0–P5ni ulangan Ubuntuda bajaring. P6da CloudShell va Ubuntu amallari uchun joy har bir bosqichda ko‘rsatiladi.

0. Umumiy qoidalar

P0–P6 mashq uchun. P0 — OSni kuzatishga bag‘ishlangan boshlang‘ich qadamma-qadam mashq. P1–P6 M1–M6dan boshqa `directory`, `user`, `group`, `service` va `port`lardan foydalanadi. M7 umumlashtiruvchi topshiriq; unga alohida mashq yo‘q.

Har bir Pda avval topshiriq matni, keyin yechim namunasi beriladi. Topshiriqni avval mustaqil bajarish yoki yo‘riqnoma bilan birga ishlash mumkin. P va M alohida baholanadi: Pdagi `PASS` Mni avtomatik `PASS` qilmaydi.

Ubuntuda quyidagini bajaring. Natija `ssm-user` bo‘lsin:

```
id -un
```

Boshqa `user` ko‘rinsa, bir marta `exit` qiling va qayta tekshiring:

```
exit
id -un
```

Har bir bosqich boshida ish `directory`siga o‘ting. O‘tgach `pwd` bilan joyni tekshiring.

Ubuntu tomonidagi baholashni `ssm-user` sifatida bajaring. P6ning CloudShell tomonini CloudShellda tekshiring:

```
jdu-check P1
```

Baholash yakuniy holatni tekshiradi. Faqat kuzatish yoki oraliq amal har doim baholanmaydi. Natija o‘qituvchi progress sahifasiga avtomatik yuboriladi. Yuborish ishlamasa ham Ubuntu yoki CloudShelldagi

PASS/FAIL ko‘rinadi.

Faqat boshidan boshlash zarur bo‘lsa reset qiling:

```
jdu-reset P1
```

P0 Environment and OS

O‘rganiladigan ko‘nikmalar

- Ubuntu OS haqidagi ma’lumot bilan Linux kernel haqidagi ma’lumotni farqlash.
- Hozirgi user, hostname va PID 1 processini haqiqiy serverda aniqlash.
- Command natijalarini filega yozish, saqlash, tekshirish va topshirish.

Topshiriq

O‘z Ubuntu mashq serveringizdan oltita qiymatni aniqlang va ~/jdu-lab/p0/observation.envga yozing. Bu eski M0 o‘rniga qadamma-qadam P0 mashqidir. Qiymatlarni misoldan yoki boshqa talabadan ko‘chirmang; faqat o‘z serveringizdagi natijalarni oling. Oltita bandni jdu-check P0 bilan tekshiring.

Yechim namunasi (qadamma-qadam)

1-bosqich: Server va userni tekshirish

Hujjat boshida aytilganidek, CloudShellda ssh jdu-ubuntuni bajaring. Quyidagi barcha amallar ulangan Ubuntuda bajariladi.

```
# Home directoryga o‘ting.
cd ~
# Joriy directoryni ko‘rsating.
pwd
# Joriy user nomini ko‘rsating.
id -un
```

id -un natijasi ssm-user bo‘lishi kerak. Bu keyin yoziladigan USER_NAME qiymati. Keyingi natijalarni o‘zingiz uchun qayd eting.

2-bosqich: OS turi va versiyasini aniqlash

```
# OS turi va versiyasini ko‘rsating.
cat /etc/os-release
```

ID= qiymatini OS_ID, VERSION_ID= qiymatini OS_VERSION_ID sifatida yozing. Bular NAME va PRETTY_NAMEdan boshqa maydonlar. Agar natijada qo‘shtirnoq bo‘lsa, faqat qiymatni yozing.

3-bosqich: Linux kernel versiyasini aniqlash

```
# Linux kernel release raqamini ko‘rsating.
uname -r
```

-r kernel releaseni ko‘rsatadi. Chiqqan butun satrni KERNEL_RELEASE sifatida qayd eting. OS versiyasi va kernel versiyasi bir xil ma’lumot emas.

4-bosqich: PID 1 process nomini aniqlash

```
# PID 1 process nomini ko‘rsating.
cat /proc/1/comm
```

/procda ishlayotgan tizim haqidagi ma’lumotlar mavjud. 1 — process ID (PID), comm esa shu process nomini ko‘rsatadigan file. Natijani PID1_COMM deb yozing.

5-bosqich: User va hostname qiymatlarini tekshirish

```
# Joriy user nomini ko‘rsating.
id -un
# Server hostname qiymatini ko‘rsating.
hostname
```

Birinchi natija USER_NAME, ikkinchisi HOST_NAME. Hostname serverni bildiradi; u user nomi yoki natija yuborishga ishlatiladigan Server ID emas.

6-bosqich: Kuzatuv fileini yaratish

```
# Home directoryga o‘ting.
cd ~
# P0 ish directorysini yarating.
mkdir -p jdu-lab/p0
# P0 ish directorysiga o‘ting.
cd jdu-lab/p0
# Joriy directoryni ko‘rsating.
pwd
# Kuzatuv fileini tahrirlang.
nano observation.env
```

nano ochilgach, quyidagi olti satrni kiriting. Har bir = belgisidan o‘ng tomonga o‘z serveringizda aniqlagan qiymatni yozing. Bu file mazmuni; uni command sifatida bajarmang:

```
OS_ID=
OS_VERSION_ID=
KERNEL_RELEASE=
PID1_COMM=
USER_NAME=
HOST_NAME=
```

Har bir nom bitta satrda bo‘lsin. = atrofida bo‘sh joy va qiymat atrofida qo‘shtirnoq bo‘lmasin. Ctrl+O bilan saqlang, observation.env nomini Enter bilan tasdiqlang, so‘ng Ctrl+X bilan chiqing.

7-bosqich: Saqlangan fileni tekshirish

```
# Joriy directoryni ko‘rsating.
pwd
# Kuzatuv filei mavjudligi va permissionini tekshiring.
ls -l observation.env
# Yozib olingan oltita maydonni ko‘rsating.
cat observation.env
```

Olti satrning hammasida 1–5-bosqichlarda olingan qiymatlar bo‘lsin. Bo‘sh yoki noto‘g‘ri qiymatni nano observation.env bilan tuzatib, qayta saqlang.

8-bosqich: Baholash va topshirish

```
# Baholash qaysi user bilan bajarilishini tekshiring.
id -un
# P0 bajarilganini baholang.
jdu-check P0
```

Oltita PASS va 6 / 6 checks cleared P0 tugaganini bildiradi. FAIL bo‘lsa, ko‘rsatilgan bandni o‘z serveringizda qayta aniqlang, fileni tuzating va baholashni takrorlang.

RESULT — topshiriq bahosi, REPORT — o‘qituvchiga yuborish natijasi. Yuborish ishlamasa, fileni o‘chirmang va mashqni reset qilmang. Ulanish hamda ro‘yxatdan o‘tishni tekshirib, yana urinib ko‘ring.

9-bosqich: Keyingi mashqqa o‘tish

```
# Keyingi mashqdan oldin home directoryga qayting.
cd ~
```

P1ga o‘ting. CloudShellga qaytmoqchi bo‘lsangizgina exitni bajaring. P0ni birinchi marta bajarishda reset kerak emas. Yozuvni o‘chirib qayta boshlash kerak bo‘lsagina Ubuntuda jdu-reset P0ni bajaring.

P1 Shell, path, file, text

O‘rganiladigan ko‘nikmalar

- home directory va ish directorysini ajratish.
- directory yaratish.
- filedan nusxa olish.
- logdan kerakli satrlarni saqlash.

Topshiriq

~/jdu-lab/p1/inboxda sozlama filei va log bor. ~/jdu-lab/p1/practice01 hali tayyor emas, unda keraksiz .tmp filelar qolgan. Quyidagi holatga keltiring:

1. practice01 ichida config, logs, notes directorylarini yarating.
2. inbox/config/training.confni practice01/config/training.confga, inbox/logs/practice.logni practice01/logs/practice.logga ko‘chiring. Manba va nusxa mazmunini o‘zgartirmang.
3. practice01 ostidagi barcha .tmp filelarni o‘chiring. Boshqa filelarni o‘chirmang.
4. Tayyor practice01 ichidagi directory va filelarning egasi Ubuntu boshqaruv useri ssm-user bo‘lsin.
5. Nusxa olingan logdagi WARN bor satrlarnigina satr raqamisiz practice01/notes/warnings.txtga yozing.
6. Shu logning oxirgi to‘rt satrini asl tartibda practice01/notes/recent.txtga yozing.

Tayyor holatni jdu-check P1 bilan tekshiring. Olti bandning hammasi PASS bo‘lsa, mashq tugaydi.

Yechim namunasi (qadamma-qadam)

1-bosqich: Home directoryga o‘tish

```
# Home directoryga o‘ting.
cd ~
# Joriy directoryni ko‘rsating.
pwd
```

Natija `/home/ssm-user` bo‘lsin.

2-bosqich: P1 directorysiga o‘tish

```
# P1 ish directorysiga o‘ting.
cd ~/jdu-lab/p1
# Joriy directoryni ko‘rsating.
pwd
# Asl filelar va tugallanmagan topshirish joyini daraxt shaklida ko‘rsating.
tree
```

`inbox` va hali tugallanmagan `practice01`ni toping.

3-bosqich: Kerakli directorylarni yaratish

```
# P1 ish directorysiga o‘ting.
cd ~/jdu-lab/p1
# Config file uchun directory yarating.
mkdir -p practice01/config
# Log uchun directory yarating.
mkdir -p practice01/logs
# Natija filei uchun directory yarating.
mkdir -p practice01/notes
# Yaratilgan tuzilishni ko‘rsating.
tree practice01
```

4-bosqich: Config filedan nusxa olish

Nusxa joylashtiriladigan `directory`ga o‘ting.

```
# Config nusxasi joylashgan directoryga o‘ting.
cd ~/jdu-lab/p1/practice01/config
# Joriy directoryni tekshiring.
pwd
# Asl config fileni joriy directoryga nusxalang.
cp ../../inbox/config/training.conf .
# Nusxalangan fileni ko‘rsating.
ls -l
```

`.` hozirgi `directory`ni anglatadi.

5-bosqich: Log filedan nusxa olish

```
# Log nusxasi joylashgan directoryga o‘ting.
cd ~/jdu-lab/p1/practice01/logs
# Joriy directoryni tekshiring.
pwd
# Asl logni joriy directoryga nusxalang.
cp ../../inbox/logs/practice.log .
# Nusxalangan fileni ko‘rsating.
ls -l
```

6-bosqich: Vaqtinchalik filelarni topib o‘chirish

practice01ga o‘ting va o‘chirishdan oldin tuzilmani ko‘ring:

```
# Yakunlash kerak bo‘lgan directoryga o‘ting.
cd ~/jdu-lab/p1/practice01
# Joriy directoryni tekshiring.
pwd
# O‘chirishdan oldingi file tuzilishini ko‘rsating.
tree
```

staging ichidan training.conf.tmp va practice.log.tmpni toping. So‘ng shu ikki fileni shart bilan qidiring:

```
# Joriy directory va uning ichidagi faqat .tmp filelarni qidiring.
find . -type f -name '*.tmp' -print
```

. — qidirish boshlanadigan joy; -type f faqat oddiy filelarni, -name '*.tmp' nomi .tmp bilan tugaydiganlarni tanlaydi; -print topilgan pathlarni ko‘rsatadi. Qo‘shtirnoq *.tmpni shell oldindan kengaytirib yubormasligi uchun kerak. Natijani tree bilan solishtiring. Faqat ikki keraksiz file ekanini aniqlagach, aynan ularning pathini ko‘rsatib o‘chiring:

```
# Tekshirilgan ikkita keraksiz vaqtinchalik fileni o‘chiring.
rm staging/training.conf.tmp staging/practice.log.tmp
# O‘chirishdan keyingi tuzilishni tekshiring.
tree
```

Ikki .tmp ko‘rinmaydi, lekin config/training.conf va logs/practice.log saqlanadi.

7-bosqich: WARN bor satrlarni saqlash

Natija yoziladigan notesga o‘ting:

```
# Natija filei directorysiga o‘ting.
cd ~/jdu-lab/p1/practice01/notes
# Joriy directoryni tekshiring.
pwd
# Logdagi WARN bor satrlarni ekranda ko‘rsating.
grep 'WARN' ../logs/practice.log
```

Faqat WARN bor satrlar ko‘ringanini tekshiring. Endi > qo‘shib, shu natijani filega yozing. > natijani ekrandan filega yo‘naltiradi; mavjud mazmunni qayta yozadi:

```
# WARN satrlarini warnings.txt fileiga saqlang.
grep 'WARN' ../logs/practice.log > warnings.txt
# Saqlangan natijani ko‘rsating.
cat warnings.txt
```

Saqlangan satrlarni oldin ekranda ko‘ringan satrlar bilan solishtiring.

8-bosqich: Oxirgi to‘rt satrni saqlash

```
# Natija filei directorysiga o‘ting.
cd ~/jdu-lab/p1/practice01/notes
# Logning oxirgi to‘rt satrini ekranda ko‘rsating.
tail -n 4 ../logs/practice.log
```

Oxirgi to‘rt satr asl tartibda ko‘ringanini tekshiring. So‘ng aynan shu natijani saqlang:

```
# Oxirgi to‘rt satrni recent.txt fileiga saqlang.
tail -n 4 ../logs/practice.log > recent.txt
# Saqlangan natijani ko‘rsating.
cat recent.txt
```

Saqlangan to‘rt satrni `cat` bilan tekshiring.

9-bosqich: Barcha natijani tekshirish

```
# P1 ish directorysiga qayting.
cd ~/jdu-lab/p1
# Tayyor directory tuzilishini ko‘rsating.
tree practice01
# P1 bajarilganini baholang.
jdu-check P1
```

Barcha 6 band `PASS` bo‘lsa, M1ga o‘ting.

P2 User, group, permission, setgid

O‘rganiladigan ko‘nikmalar

- `primary group` va `supplementary group`ni tekshirish.
- Umumiy `directory`ga `group permission` qo‘yish.
- `setgid` yordamida yangi `file`ning `group`ini meros qildirish.

Topshiriq

`/srv/jdu-practice-share` umumiy `directory`sida yozuvchi va o‘quvchi huquqlarini ajrating. Dastlab `jdupracticeviewer` noto‘g‘ri ravishda `practiceops supplementary group`iga kirgan, `jdupracticewriter` esa kirmagan. `Directory` va `GUIDE.txt`ning egasi, `group`i va `permission`i ham tayyor emas.

1. `jdupracticewriter`ni `practiceops supplementary group`iga qo‘shing, `jdupracticeviewer`ni undan chiqaring.
2. `/srv/jdu-practice-share` egasi va `group`ini `root:practiceops`, `mode`ini `2775` qiling. Yangi `file`lar `directory` `group`ini meros qilsin.
3. `GUIDE.txt` egasi va `group`ini `root:practiceops`, `mode`ini `664` qiling.
4. `ssm-user`dan `jdupracticewriter`ga o‘tib, umumiy `directory`da `writer-created.txt` yarating. Yangi `file` `group`i `practiceops` ekanini tekshiring.
5. `ssm-user`ga qaytib, `jdupracticeviewer`ga o‘ting. `GUIDE.txt`ni o‘qish mumkin, lekin umumiy `directory`da yangi `file` yaratish mumkin emasligini tekshiring.

Boshqa `user` bilan amallar tugagach, `ssm-user`ga qayting va `jdu-check P2`ni bajaring. Oltita baho `group` a’zoligi, `directory`, `file`, `group` merosi va o‘quvchi huquqiga tegishli.

Baholash yakuniy holatni tekshiradi. 4-bandda yaratilgan `writer-created.txt` kuzatish uchun kerak; uning nomi topshiriladigan javob sifatida tekshirilmaydi. `jdu-check` baholash vaqtida writer nomidan vaqtinchalik `file` yaratib, uning `group`i `practiceops` ekanini tekshiradi. Viewer huquqi faqat shu yaratish tekshiruvi muvaffaqiyatli bo‘lgandan keyin `PASS` bo‘ladi. Shu sabab yaratish tekshiruvi muvaffaqiyatsiz bo‘lsa, viewer huquqi to‘g‘ri bo‘lsa ham oxirgi ikkita band `FAIL` bo‘ladi.

Tekshiruv ID	Talab qilinadigan yakuniy holat
P2-ID-01	jdupracticewriter practiceops a’zosi, jdupracticeviewer esa a’zo emas.
P2-PERM-01	Umumiy directory egasi va groupi root:practiceops.
P2-PERM-02	Umumiy directory modei 2775, shu jumladan setgid.
P2-PERM-03	GUIDE.txt egasi, groupi va modei root:practiceops, 664.
P2-PERM-04	Writer yangi file yarata oladi; uning groupi practiceops.
P2-PERM-05	Yuqoridagi yaratish tekshiruvi muvaffaqiyatli; viewer GUIDE.txtni o‘qiy oladi, ammo yangi file yarata olmaydi.

Birinchi band FAIL bo‘lsa, 3-bosqichdan keyingi id jdupracticewriter va id jdupracticeviewer natijalarini tekshiring. Oxirgi ikki band FAIL bo‘lsa, avval P2-PERM-04 sababini aniqlang. jdu-checkni albatta ssm-user sifatida bajaring. Sozlamalar to‘g‘ri bo‘lsa ham FAIL qolsa, reset qilmang; FAIL bo‘lgan tekshiruv IDlarini o‘qituvchiga ayting.

Yechim namunasi (qadamma-qadam)

1-bosqich: Boshqaruv userini tekshirish

```
# Home directoryga o‘ting.
cd ~
# Joriy user nomini ko‘rsating.
id -un
# Joriy user ID va tegishli grouplarni ko‘rsating.
id
```

2-bosqich: Boshlang‘ich groupni tekshirish

```
# Umumiy group yozuvini ko‘rsating.
getent group practiceops
# Writer user tegishli grouplarni tekshiring.
id jdupracticewriter
# Viewer user tegishli grouplarni tekshiring.
id jdupracticeviewer
```

Dastlab jdupracticeviewer practiceops a’zosi, jdupracticewriter esa a’zo emas.

3-bosqich: Group a’zoligini to‘g‘rilash

```
# Home directoryga o‘ting.
cd ~
# Writer userni umumiy groupga qo‘shing.
sudo usermod -aG practiceops jdupracticewriter
# Viewer userni umumiy groupdan chiqaring.
sudo gpasswd -d jdupracticeviewer practiceops
# O‘zgarishdan keyingi umumiy groupni ko‘rsating.
getent group practiceops
# Writer userning group a’zoligini tekshiring.
id jdupracticewriter
# Viewer userning group a’zoligini tekshiring.
id jdupracticeviewer
```

4-bosqich: Umumiy directoryni sozlash

```
# /srv directorysiga o‘ting.
cd /srv
# Joriy directoryni ko‘rsating.
pwd
# Umumiy directoryning boshlang‘ich permissionini ko‘rsating.
ls -ld jdu-practice-share
```

O‘zgartirishdan oldin directoryning permissioni, egasi va groupini ko‘ring. ls -lddagi -d directoryning ichidagilarni emas, aynan o‘zini ko‘rsatadi. Avval permission, havola sonidan keyin esa ega va group chiqadi.

```
# Umumiy directory owner va group qiymatini o‘zgartiring.
sudo chown root:practiceops jdu-practice-share
# Owner va group o‘zgarganini tekshiring.
ls -ld jdu-practice-share
# Group merosini o‘z ichiga olgan 2775 permissionni o‘rnating.
sudo chmod 2775 jdu-practice-share
# setgid uchun s belgisi va permission qiymatini tekshiring.
ls -ld jdu-practice-share
```

chowndan keyin ega va group root practiceops, chmoddan keyin permission drwxrwsr-x bo‘lsin. Groupning bajarish joyidagi s — setgid; shu directoryda yaratilgan yangi file uning groupini oladi.

5-bosqich: GUIDE.txtni sozlash

Kerakli directoryga o‘ting:

```
# Umumiy directoryga o‘ting.
cd /srv/jdu-practice-share
# Joriy directoryni ko‘rsating.
pwd
# GUIDE.txtning boshlang‘ich permission qiymatini ko‘rsating.
ls -l GUIDE.txt
```

Avval filening boshlang‘ich permissioni, egasi va groupini ko‘ring:

```
# GUIDE.txt owner va group qiymatlarini o‘zgartiring.
sudo chown root:practiceops GUIDE.txt
# Owner va groupni tekshiring.
ls -l GUIDE.txt
# Owner va groupga o‘qish hamda yozish huquqini bering.
sudo chmod 664 GUIDE.txt
# Filening yakuniy permissionini tekshiring.
ls -l GUIDE.txt
```

Yakuniy natija root practiceops va -rw-rw-r--. Ega va group o‘qib-yozadi, boshqa userlar faqat o‘qiydi.

6-bosqich: Writerga o‘tib file yaratish

```
# Home directoryga qayting.
cd ~
# Writer user shelliga o‘ting.
sudo su - jdupracticewriter
# Almashgandan keyingi user nomini tekshiring.
id -un
# Almashgandan keyingi grouplarni tekshiring.
id
# Umumiy directoryga o‘ting.
cd /srv/jdu-practice-share
# Joriy directoryni tekshiring.
pwd
# Writer user sifatida file yarating.
printf '%s\n' 'guided writer file' > writer-created.txt
# Yaratilgan filening owner groupini tekshiring.
ls -l writer-created.txt
# Writer user shellidan chiqing.
exit
# Boshqaruvchi userga qaytganingizni tekshiring.
id -un
```

Oxirgi natija `ssm-user` bo‘lsin.

7-bosqich: Viewerning o‘qish va yozishini tekshirish

```
# Home directoryga qayting.
cd ~
# Viewer user shelliga o‘ting.
sudo su - jdupracticeviewer
# Almashgandan keyingi user nomini tekshiring.
id -un
# Viewer user tegishli grouplarni tekshiring.
id
# Umumiy directoryga o‘ting.
cd /srv/jdu-practice-share
# Joriy directoryni tekshiring.
pwd
# GUIDE.txtni viewer user sifatida o‘qing.
cat GUIDE.txt
# Yozish rad etilishini tekshiring.
touch viewer-created.txt
# Viewer user shellidan chiqing.
exit
# Boshqaruvchi userga qaytganingizni tekshiring.
id -un
```

`cat` muvaffaqiyatli ishlaydi. `touch` `Permission denied` beradi. Bu kutilgan rad etish.

`jdupracticeviewer` `root` egasi emas va `practiceops`dan chiqarilgan, shuning uchun unga «boshqalar» permissioni qo‘llanadi. `GUIDE.txt`da bu o‘qishga (`r--`) ruxsat beradi. `Directory`da esa `r-x` bor, lekin yozish (`w`) yo‘q. Yangi `file` yaratish uchun `directory`da yozish va o‘tish (`x`) huquqi kerak; shu sabab `touch` rad etiladi.

8-bosqich: Baholash

```
# Home directoryga qayting.
cd ~
# Baholashdan oldin user nomini tekshiring.
id -un
# P2 bajarilganini baholang.
jdu-check P2
```

Barcha 6 band PASS bo‘lsa, M2ga o‘ting.

P3 Process va package

8-bobdan keyin 1–4-bosqichlarda processni tekshirib, serviceni to‘xtating. 9-bobni o‘qib, 5-bosqichda packageni o‘rnating.

O‘rganiladigan ko‘nikmalar

- systemd service va process PIDini bog‘lash.
- Main PIDni ps bilan tekshirish va kerakli serviceni to‘xtatish.
- apt bilan package o‘rnatish.

Topshiriq

Boshlang‘ich holatda jdu-p3-process1.service, jdu-p3-process2.service va jdu-p3-process3.service ishlayapti. figlet packagei o‘rnatilmagan. A va Bni bajaring.

Topshiriq A — faqat process2ni to‘xtatish. jdu-p3-process2.service Main PIDini aniqlang va shu PIDdagi processni ps bilan tekshiring. systemctl stop orqali serviceni to‘xtating. Faqat process2 to‘xtasin; process1 va process3 ishlashda davom etsin.

Topshiriq B — package o‘rnatish. figlet ma’lumotini ko‘ring, apt bilan o‘rnating va commandni bir marta bajaring.

jdu-check P3 A va Bni bittadan band sifatida baholaydi, jami ikkita. Command history yoki yozgan eslatmangiz topshirilmaydi.

Yechim namunasi (qadamma-qadam)

1-bosqich: Uchta serviceni tekshirish

```
# Home directoryga o‘ting.
cd ~
# P3 servicelar ro‘yxatini ko‘rsating.
systemctl list-units --type=service 'jdu-p3-*'
# process1 holatini ko‘rsating.
systemctl status jdu-p3-process1.service --no-pager
# process3 holatini ko‘rsating.
systemctl status jdu-p3-process3.service --no-pager
```

2-bosqich: Process2 Main PIDini o‘qish

```
# process2 holati va Main PID qiymatini ko‘rsating.
systemctl status jdu-p3-process2.service
```

Main PID: o‘ngidagi raqamni o‘zingiz uchun yozib oling. Pager ochilsa, q bilan chiqing.

3-bosqich: Yozib olingan PIDdagi processni tekshirish

Masalan, yozib olgan raqam 1234 bo‘lsa, quyidagi commandda 1234ni o‘z raqamingizga almashtiring:

```
# Yozib olgan PID processini ko‘rsating; 1234 o‘rniga o‘z qiymatingizni qo‘ying.
ps -fp 1234
```

PID, USER va CMDni tekshiring.

4-bosqich: Process2 serviceni to‘xtatish

```
# process2 serviceni to‘xtating.
sudo systemctl stop jdu-p3-process2.service
# process1 ishlayotganini tekshiring.
systemctl is-active jdu-p3-process1.service
# process2 to‘xtaganini tekshiring.
systemctl is-active jdu-p3-process2.service
# process3 ishlayotganini tekshiring.
systemctl is-active jdu-p3-process3.service
```

Tartib bilan active, inactive, active chiqishi kerak.

5-bosqich: Figletni tekshirib o‘rnatish

```
# Home directoryga o‘ting.
cd ~
# figlet package ma’lumotini o‘qing.
apt show figlet
# Mavjud packagelar ro‘yxatini yangilang.
sudo apt update
# figletni o‘rnating.
sudo apt install -y figlet
# figlet command joylashuvini tekshiring.
command -v figlet
# figletni bir marta ishga tushiring.
figlet JDU
# P3 bajarilganini baholang.
jdu-check P3
```

figlet JDU yozuv chiqarib, o‘zi tugaydi. jdu-check P3 ikkala topshiriqni bittadan band sifatida baholaydi.

Ikkala band PASS bo‘lsa, M3ga o‘ting.

P4 systemd service

O‘rganiladigan ko‘nikmalar

- unit fileni o‘qish.
- active va enabled holatlarini alohida sozlash.
- Unit sozlamalaridagi User, WorkingDirectory va ExecStartni o‘qish.

Topshiriq

jdu-practice-status.service unit filei tayyor, lekin service to‘xtagan va avtomatik ishga tushish o‘chirilgan. Unit fileni o‘zgartirmasdan:

1. User, WorkingDirectory va ExecStartni o‘qing.
2. Serviceni ishga tushiring va hozir active ekanini tekshiring.
3. Avtomatik ishga tushishni yoqing va enabled ekanini tekshiring. Bu activedan alohida holat.

jdu-check P4 o‘zgarmagan unitning ishlashi va avtomatik ishga tushishini ikkita bandda baholaydi. Kuzatuv natijasini alohida file sifatida topshirmang.

Yechim namunasi (qadamma-qadam)

1-bosqich: Unit fileni o‘qish

```
# Home directoryga o‘ting.
cd ~
# Service unit sozlamasini ko‘rsating.
systemctl cat jdu-practice-status.service
```

User, WorkingDirectory, ExecStartni tekshiring.

2-bosqich: Dastlabki holatni tekshirish

```
# Service hozir ishlayotganini tekshiring.
systemctl is-active jdu-practice-status.service
# Boot paytidagi avtomatik ishga tushish yoqilganini tekshiring.
systemctl is-enabled jdu-practice-status.service
```

3-bosqich: Serviceni start qilish

```
# Serviceni hozir ishga tushiring.
sudo systemctl start jdu-practice-status.service
# Servicening joriy holatini tekshiring.
systemctl is-active jdu-practice-status.service
```

4-bosqich: Tizim yoqilganda start qilishni sozlash

```
# Boot paytidagi avtomatik ishga tushishni yoqing.
sudo systemctl enable jdu-practice-status.service
# Avtomatik ishga tushish sozlamasini tekshiring.
systemctl is-enabled jdu-practice-status.service
```

5-bosqich: Baholash

```
# Home directoryga o‘ting.
cd ~
# P4 bajarilganini baholang.
jdu-check P4
```

Ikkala band PASS bo‘lsa, M4ga o‘ting.

P5 Port, socket, HTTP, journal

O‘rganiladigan ko‘nikmalar

- service, process va listening socketni bog‘lash.
- IP address va portni o‘qish.
- HTTP request journalda qayd etilishini tekshirish.

Topshiriq

jdu-practice-web.service to‘xtagan. Unit fileni o‘zgartirmang. Quyidagi bog‘lanishni yarating va tekshiring:

1. Unit filedan service user, chiqariladigan file path, IP address va portni aniqlang. Serviceni ishga tushiring.

2. 127.0.0.1:8181da kutayotgan socketni toping. Uning PIDini service Main PIDi bilan solishtiring. Barcha manzillarda kutishga o‘zgartirmang.
3. http://127.0.0.1:8181/ga request yuborib, HTTP 200 va javob mazmunini ko‘ring. Service user /srv/jdu-practice-web/index.txtni o‘qiy olishini ham tekshiring.
4. http://127.0.0.1:8181/p5-checkga o‘zingiz request yuboring. Servicening joriy ishga tushishiga tegishli journalda REQUEST path=/p5-check yozuvi paydo bo‘lsin.
5. 18181 portida listener yo‘qligini tekshirib, ishlayotgan 8181 bilan solishtiring.

jdu-check P5 to‘rtta bandni baholaydi: socket, HTTP va file access, request logi hamda ochiq-yopiq portlar farqi.

Yechim namunasi (qadamma-qadam)

1-bosqich: Unit fileni o‘qish

```
# Home directoryga o‘ting.
cd ~
# Web service unit sozlamasini ko‘rsating.
systemctl cat jdu-practice-web.service
```

127.0.0.1, 8181, jdupracticeweb va kontent filei pathini tekshiring.

2-bosqich: Serviceni start qilish

```
# Web serviceni hozir ishga tushiring.
sudo systemctl start jdu-practice-web.service
# Service ishlayotganini tekshiring.
systemctl is-active jdu-practice-web.service
```

3-bosqich: Listening socketni tekshirish

```
# 8181-portning listening address va PID qiymatini ko‘rsating.
sudo ss -lntp | grep ':8181'
```

127.0.0.1:8181ni toping. Shu chiqishda 0.0.0.0:8181 yoki [::]:8181da kutish yo‘qligini ham tekshiring. Birinchisi faqat shu host ichidan ulanishni, keyingilari barcha manzillarda kutishni bildiradi.

4-bosqich: Socket PID va Main PIDni solishtirish

```
# Service holatini ko‘rsating va Main PID raqamini o‘qing.
systemctl status jdu-practice-web.service --no-pager
# 8181-portni tinglayotgan processning pid= qiymatini o‘qing.
sudo ss -lntp | grep ':8181'
```

systemctl statusdagi Main PID: raqamini ss natijasining users: qatoridagi pid= bilan solishtiring. Teng bo‘lsa, servicening asosiy processi 8181 portida kutmoqda.

5-bosqich: HTTP responseni tekshirish

```
# HTTP response header va body qismlarini ko‘rsating.
curl -i http://127.0.0.1:8181/
```

6-bosqich: Service userning o‘qish huquqini tekshirish

Kontent directorysiga o‘ting:

```
# Ochiq file joylashgan directoryga o‘ting.
cd /srv/jdu-practice-web
# Joriy directoryni ko‘rsating.
pwd
# File yo‘lidagi har bir directory permissionini ko‘rsating.
ls -ld /srv /srv/jdu-practice-web
# Ochiq filening owner, group va permission qiymatlarini ko‘rsating.
ls -l index.txt
# Service user tegishli grouplarni ko‘rsating.
id jdupracticeweb
# Service userning login shell qiymatini ko‘rsating.
getent passwd jdupracticeweb
# Fileni service user huquqi bilan faqat shu buyruq uchun o‘qing.
sudo -u jdupracticeweb -- cat index.txt
```

`ls -ld` har bir directoryning, `ls -l` esa index.txtning egasi, groupi va permissionini ko‘rsatadi. Fileni o‘qish uchun uning read permissioni va har bir yuqori directorydan o‘tish (`x`) huquqi kerak. `id` service user grouplarini, `getent passwd` uning login shellini ko‘rsatadi.

`sudo -u jdupracticeweb -- cat index.txt` catni faqat bir marta jdupracticeweb huquqi bilan bajaradi. `--` sudo opsiyalari tugaganini bildiradi. Mazmun chiqsa, o‘qish mumkin. Bu service userning shelli /usr/sbin/nologin; shuning uchun interaktiv `su -` ishlatmang.

7-bosqich: Talaba requestini yuborish

```
# Ko‘rsatilgan pathga HTTP request yuboring.
curl -i http://127.0.0.1:8181/p5-check
# Servicening so‘nggi journal yozuvlarini ko‘rsating.
sudo journalctl -u jdu-practice-web.service --no-pager -n 20
```

`REQUEST path=/p5-check`ni toping.

8-bosqich: Closed port bilan solishtirish

```
# Listening socket yo‘q 18181-portga ulanishni sinang.
curl --max-time 2 http://127.0.0.1:18181/
# 18181-portda listening socket yo‘qligini tekshiring.
sudo ss -lnt | grep ':18181'
```

Ikkalasi ham muvaffaqiyatli chiqmaydi. TCP 18181da kutayotgan `socket` yo‘q.

9-bosqich: Baholash

```
# Home directoryga o‘ting.
cd ~
# P5 bajarilganini baholang.
jdu-check P5
```

To‘rttala band `PASS` bo‘lsa, M5ga o‘ting.

P6 SSH, remote command, scp

O‘rganiladigan ko‘nikmalar

- CloudShell va Ubuntuni farqlash.
- `scp` bilan `upload` va `download` qilish.

- SSH remote commandini bajarish.

Topshiriq

CloudShellni ulanish boshlanadigan tomon, Ubuntuni esa uzoqdagi tomon deb oling. SSH sozlamasi va jdu-ubuntu ulanish nomi tayyor. P6 filelari hali mavjud emas.

Uzoqdagi serverda yaratiladigan natija filei quyidagi uch satrdan iborat bo‘lsin. = o‘ngidagi qiymatlarni taxmin qilib qo‘lda yozmang; haqiqiy command natijalaridan oling:

```
REMOTE_USER=haqiqiy user nomi
REMOTE_HOST=haqiqiy hostname
REMOTE_PATH=haqiqiy home directory
```

1. CloudShelldagi ~/jdu-lab/p6/practice-source.txtga faqat JDU SSH guided transfer satrini yozing.
2. scp bilan uni Ubuntuga ~/jdu-lab/p6/practice-upload.txt sifatida yuboring. Yuborishdan oldingi va keyingi SHA-256ni solishtiring.
3. CloudShelldan SSH remote commandni bajaring. Ubuntu haqidagi haqiqiy uch qiymatni yuqoridagi shaklda ~/jdu-lab/p6/practice-remote-result.txtga yozing.
4. Ubuntuga ulanib, jdu-check P6ni bajaring. Ubuntu tomonidagi ikki band PASS bo‘lsin.
5. Natija fileini scp bilan CloudShellga ~/jdu-lab/p6/practice-downloaded-result.txt sifatida qaytaring. Ikkala tomondagi SHA-256ni solishtiring.
6. CloudShellda jdu-check P6ni bajaring. CloudShell tomonidagi to‘rt band PASS bo‘lsin.

P6 Ubuntu 2/2 va CloudShell 4/4 bo‘lganda tugaydi. O‘qituvchi progress sahifasi ham ikki tomonni alohida ko‘rsatadi.

Yechim namunasi (qadamma-qadam)

1-bosqich: Ubuntudan CloudShellga qaytish

Ubuntu promptida bajaring:

```
# Ubuntu SSH sessiyasini yakunlab, CloudShellga qayting.
exit
```

CloudShellda qayerda ekaningizni tekshiring:

```
# CloudShell home directorysiga o‘ting.
cd ~
# CloudShelldagi joriy directoryni ko‘rsating.
pwd
# CloudShell user nomini ko‘rsating.
id -un
# CloudShell hostname qiymatini ko‘rsating.
hostname
```

2-bosqich: CloudShelldagi ish directorysini yaratish

```
# CloudShellda P6 ish directorysini yarating.
mkdir -p ~/jdu-lab/p6
# CloudShelldagi P6 directorysiga o‘ting.
cd ~/jdu-lab/p6
# Joriy directoryni tekshiring.
pwd
```

3-bosqich: Upload manbasi bo‘lgan fileni yaratish

```
# Manba filega ko‘rsatilgan bitta satrni yozing.
printf '%s\n' 'JDU SSH guided transfer' > practice-source.txt
# Manba file mazmunini ko‘rsating.
cat practice-source.txt
# Ko‘chirishdan oldingi SHA-256 qiymatini ko‘rsating.
sha256sum practice-source.txt
```

4-bosqich: Ubuntudagi directoryni yaratish

CloudShellda bajaring:

```
# SSH orqali Ubuntu tomonidagi P6 directorysini yarating.
ssh jdu-ubuntu 'mkdir -p ~/jdu-lab/p6'
```

5-bosqich: Ubuntuga upload qilish

CloudShell P6 `directory`sidan bajaring:

```
# CloudShelldagi P6 directorysiga o‘ting.
cd ~/jdu-lab/p6
# scp yordamida fileni CloudShelldan Ubuntuga yuboring.
scp practice-source.txt jdu-ubuntu:~/jdu-lab/p6/practice-upload.txt
# Ubuntuda qabul qilingan filening SHA-256 qiymatini ko‘rsating.
ssh jdu-ubuntu 'sha256sum ~/jdu-lab/p6/practice-upload.txt'
# CloudShelldagi manba filening SHA-256 qiymatini ko‘rsating.
sha256sum practice-source.txt
```

Ikki SHA-256 qiymati bir xil bo‘lsin.

6-bosqich: SSH remote command bilan natija fileni yaratish

CloudShellda quyidagi bitta `command`ni aynan ko‘rsatilgandek bajaring:

```
# Ubuntudagi uchta haqiqiy qiymatni natija fileiga saqlang.
ssh jdu-ubuntu 'cd ~/jdu-lab/p6 && printf "REMOTE_USER=%s\nREMOTE_HOST=%s\nREMOTE_PATH=%s\n" "$(id -un)"
"$(hostname)" "$HOME" > practice-remote-result.txt'
```

Mazmunni `remote`da tekshiring:

```
# SSH orqali Ubuntudagi natija fileni ko‘rsating.
ssh jdu-ubuntu 'cat ~/jdu-lab/p6/practice-remote-result.txt'
```

7-bosqich: Ubuntu tomonini baholash

Ubuntuga ulanib bajaring:

```
# CloudShelldan Ubuntuga ulaning.
ssh jdu-ubuntu
# Ubuntudagi P6 directorysiga o‘ting.
cd ~/jdu-lab/p6
# Ubuntudagi joriy directoryni ko‘rsating.
pwd
# Ko‘chirilgan file va natija fileni tekshiring.
ls -l
# Ubuntu tomonidagi P6ni baholang.
jdu-check P6
# Ubuntu SSH sessiyasini yakunlab, CloudShellga qayting.
exit
```

Ubuntu tomondagi ikkala band `PASS` bo‘lsin.

8-bosqich: Natija fileni download qilish

CloudShellda bajaring:

```
# CloudShelldagi P6 directorysiga o‘ting.
cd ~/jdu-lab/p6
# scp yordamida natija fileni Ubuntudan CloudShellga qaytaring.
scp jdu-ubuntu:~/jdu-lab/p6/practice-remote-result.txt practice-downloaded-result.txt
# Yuklab olingan file mazmunini ko‘rsating.
cat practice-downloaded-result.txt
# CloudShelldagi yuklab olingan filening SHA-256 qiymatini ko‘rsating.
sha256sum practice-downloaded-result.txt
# Ubuntudagi manba filening SHA-256 qiymatini ko‘rsating.
ssh jdu-ubuntu 'sha256sum ~/jdu-lab/p6/practice-remote-result.txt'
```

Ikki SHA-256 qiymati bir xil bo‘lsin.

9-bosqich: CloudShell tomonini baholash

```
# CloudShelldagi P6 directorysiga o‘ting.
cd ~/jdu-lab/p6
# CloudShell tomonidagi P6ni baholang.
jdu-check P6
```

CloudShell tomondagi to‘rtala band `PASS` bo‘lsa, M6ga o‘ting.

O‘rganish tartibi

Kitobdagi boblar dars soatlari bilan birma-bir mos emas. Tushungan joyingizdan mashqqa o‘ting, zarur bo‘lsa bobga qayting.

- 1–3-boblar va 4-bobning tahrirlash asoslarini o‘qing. Keyin P0 qadamlari bilan haqiqiy OSni kuzating.
- 4–5-boblarni o‘qib P1, M1ga o‘ting.
- 6–7-boblarni o‘qib P2, M2ga o‘ting.
- 8-bobni o‘qib P3ning 1–4-bosqichlarini bajaring. 9-bobni o‘qib P3ning 5-bosqichini bajaring. Keyin M3ga o‘ting.
- 10-bobdan keyin P4, M4; 11-bobdan keyin P5, M5ga o‘ting.
- 12-bobdan keyin P6, M6ga o‘ting. Avvalgi ko‘nikmalarni birlashtirib M7ni bajaring.

Umumiy tavsiya etilgan tartib:

```
P0 → P1 → M1 → P2 → M2 → P3 → M3 → P4 → M4
  → P5 → M5 → P6 → M6 → M7 (yakuniy topshiriq)
```

Pda avval mustaqil urinib, keyin yechimga qarashingiz yoki boshidan namuna bo‘yicha ishlashingiz mumkin. M1–M6da P bosqichlariga qaramay, kerakli commandlarni o‘zingiz tanlang. M7da M2, M4 va M5dagi bilimlarni birlashtiring.